

Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2, Semester 1- 2026

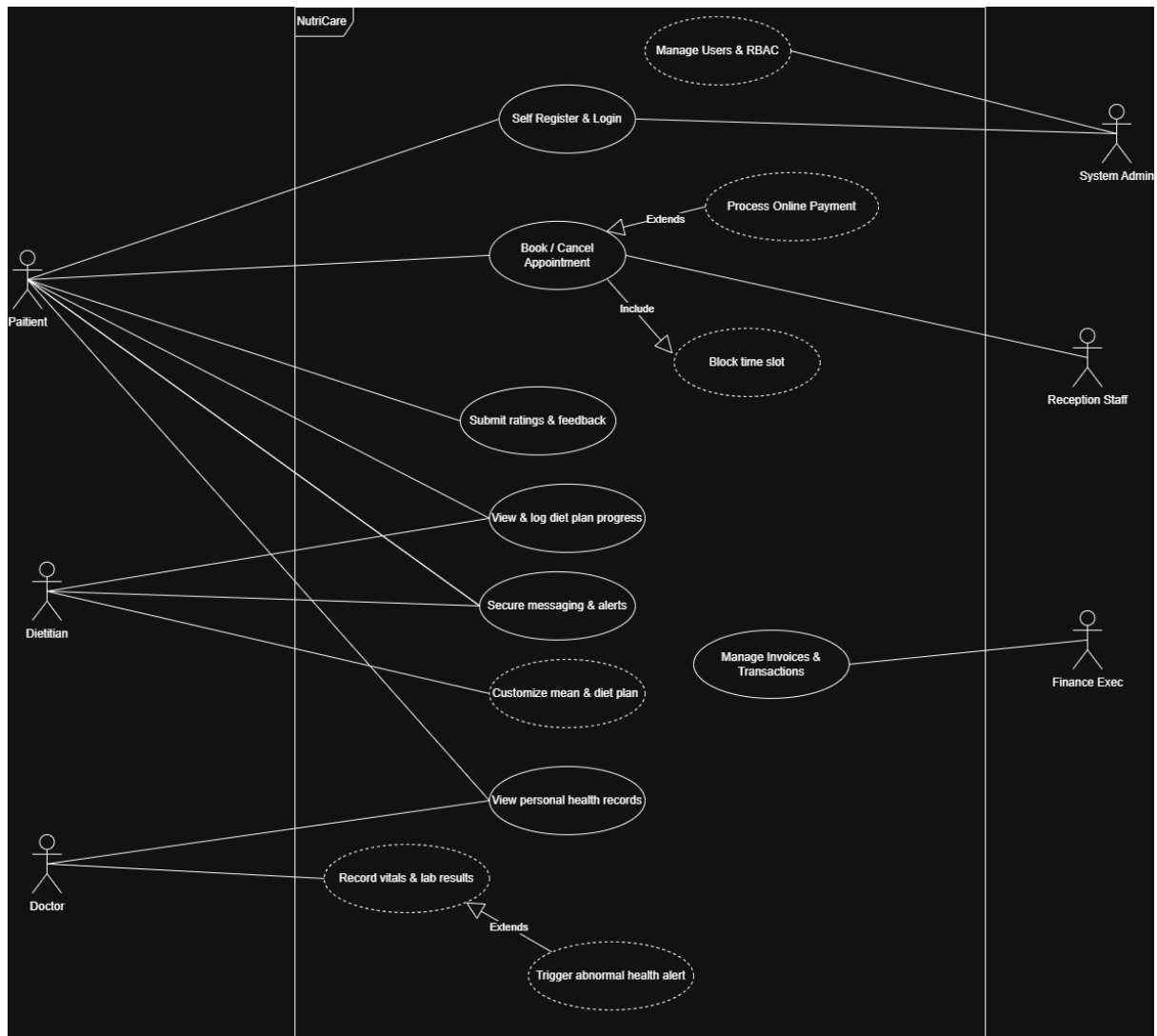
Web-based Diet Planning & Health Check-up System

Lab 03

2026-Y2-S1-MLB-B5G1-01

IT Number	Name
IT25101803	Vidanage.T.L
IT25102636	Herath H.M.H.Y
IT25103681	Hasaranga S. O
IT25101696	Mohanaranjan M.
IT25103601	Punsara M. H.
IT25100792	Gunathilaka M D S T

Use Case Diagram



Individual Use Case Scenario

Number	UC-01	
Member Assigned	Vidanage.T.L (IT25101803)	
Name	User Registration & Role-Based Access Control	
Pre-Condition	System is online and database connection is active.	
Post-Condition	A user account is created with assigned RBAC permissions.	
Primary Actor	Patient / System Admin	
Main Scenario	Step	Action
	1	Patient navigates to registration page and enters personal, contact, and medical history details.
	2	System validates user input (email format, password strength).
	3	System creates account with 'Patient' role and hashes password.
	4	Patient logs in using valid credentials.
	5	System verifies role-based permissions and routes user to Patient Dashboard.
Extension	2a. Validation failure: System alerts missing/invalid fields and prompts re-entry.	
	4a. Invalid Credentials: System displays error message and locks account after 5 failed attempts.	

Number	UC-02	
Member Assigned	Hasaranga S. O (IT25103681)	
Name	Real-Time Appointment Booking & Billing	
Pre-Condition	Patient/Receptionist is logged in Doctor/Dietitian schedule is configured.	
Post-Condition	Slot is reserved, invoice generated, and slot blocked in real time.	
Primary Actor	Patient / Receptionist	
Main Scenario	Step	Action
	1	User selects desired Practitioner (Dietitian/Doctor) and views real-time available time slots.
	2	User chooses preferred time slot.
	3	System immediately locks the slot using real-time concurrency control.
	4	System generates invoice for the selected consultation.
	5	User completes payment via Payment Gateway / cash recording.
	6	System confirms booking and sends confirmation notification via SMS/Email API.
Extension	3a. Slot simultaneously selected by another user: System notifies slot unavailability and suggests alternative times.	
	5a. Payment Failed: System releases blocked slot after 10-minute timeout.	

Number	UC-03	
Member Assigned	Herath H.M.H.Y (IT25102636)	
Name	Record Health Check-up Data & Trigger Alerts	
Pre-Condition	Patient has attended consultation. Doctor is logged in.	
Post-Condition	Health metrics logged. alerts generated if parameters exceed thresholds.	
Primary Actor	Doctor / Medical Centre Coordinator	
Main Scenario	Step	Action
	1	Doctor opens patient record and enters vital signs (BMI, BP, Blood Sugar, Temperature) and lab test results.
	2	System saves records into the MySQL Database under patient history.
	3	System compares new vital parameters against defined normal threshold limits.
	4	Parameters are identified within normal range; consultation notes are finalized.
Extension	3a. Elevated Vitals Detected (e.g., Blood Sugar > Threshold): System automatically flags record as 'High Priority Alert'. System dispatches immediate alert message to assigned Dietitian dashboard.	

Number	UC-04	
Member Assigned	Mohanaranjan M. (IT25101696)	
Name	Personalized Diet Plan Customization & Tracking	
Pre-Condition	Patient health metrics logged. Dietitian is authenticated.	
Post-Condition	Diet plan created/updated. progress tracking charts rendered.	
Primary Actor	Dietitian / Patient	
Main Scenario	Step	Action
	1	Dietitian accesses patient profile and reviews health check-up history.
	2	Dietitian sets nutritional targets and builds customized meal schedules (recommended foods, portion sizes).
	3	Dietitian publishes diet plan; system notifies patient.
	4	Patient views plan and logs daily meal completion / water intake.
	5	System updates dynamic weight and BMI visual trend charts.
Extension	2a. Condition-specific alert: System auto-suggests meal exclusion templates (e.g., low-sodium for hypertension).	

Number	UC-05	
Member Assigned	Punsara M. H. (IT25103601)	
Name	Secure Messaging & Automated Reminders	
Pre-Condition	Communication module is active. external APIs configured.	
Post-Condition	Messages delivered and logged. reminder notifications sent.	
Primary Actor	Patient / Dietitian	
Main Scenario	Step	Action
	1	Patient composes secure non-urgent message to assigned dietitian.
	2	System routes message to Dietitian's secure inbox.
	3	Dietitian reviews and responds asynchronously.
	4	System background service triggers automated appointment reminder 24 hours prior to scheduled time.
	5	Reminder is dispatched via SMS/Email API to patient.
Extension	5a. API Dispatch Failure: System logs retry status in database and re-attempts dispatch after 5 minutes.	

Number	UC-06	
Member Assigned	Gunathilaka M.D.S.T (IT25100792)	
Name	Review, Feedback & Performance Analytics	
Pre-Condition	Consultation/appointment completed.	
Post-Condition	Feedback saved. management analytics updated.	
Primary Actor	Patient / Operations Manager	
Main Scenario	Step	Action
	1	Patient receives feedback prompt post-consultation.
	2	Patient submits star rating (1-5) and written feedback/complaint.
	3	System aggregates rating into dietitian performance record.
	4	Operations Manager opens executive dashboard to view consolidated service performance analytics.
Extension	2a. Rating <= 2 stars (Low Rating / Complaint): System automatically flags review and creates a complaint ticket for Operations Manager follow-up.	